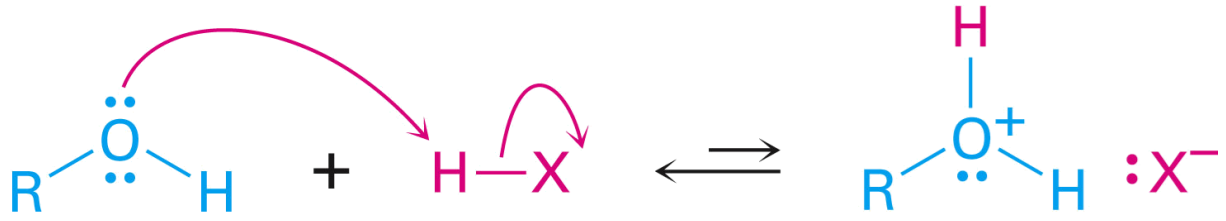
Chapter 17
Alcohols and Phenols

Properties of Alcohols and Phenols

- Weakly basic and weakly acidic
- Weak bases, protonated by strong acids yield oxonium ions, ROH_2^+

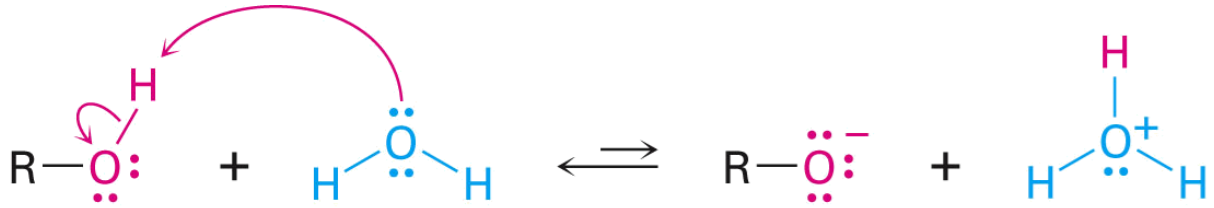


An alcohol **An oxonium ion**

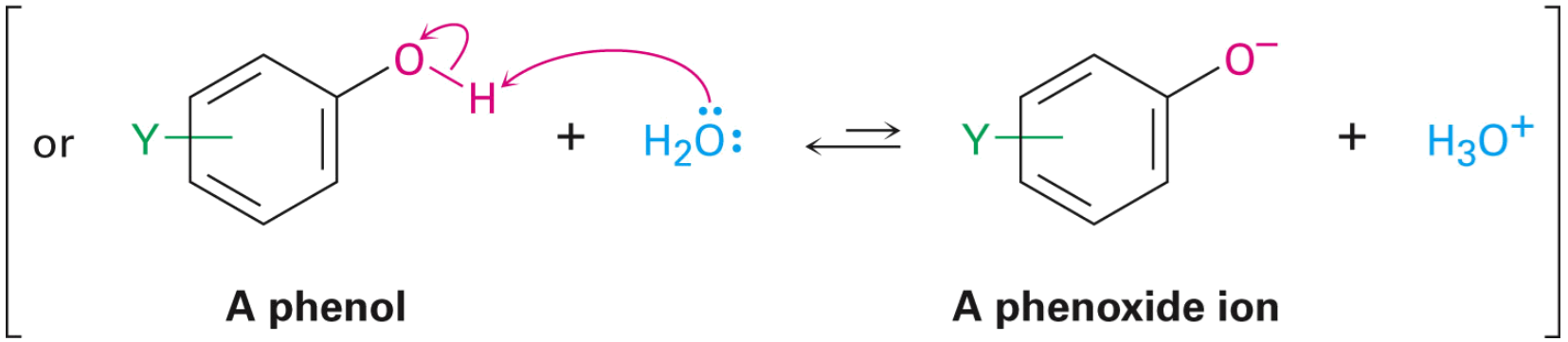
$$\left[\text{or } ArOH + HX \rightleftharpoons ArOH_2^+ \; X^-\right]$$

Properties of Alcohols and Phenols

- Weak acids, dissociate slightly in dilute aqueous solution generating H_3O^+ and an **alkoxide ion, RO^-** or **phenoxide ion, ArO^-**



An alcohol **An alkoxide ion**



A phenol **A phenoxide ion**

Properties of Alcohols and Phenols

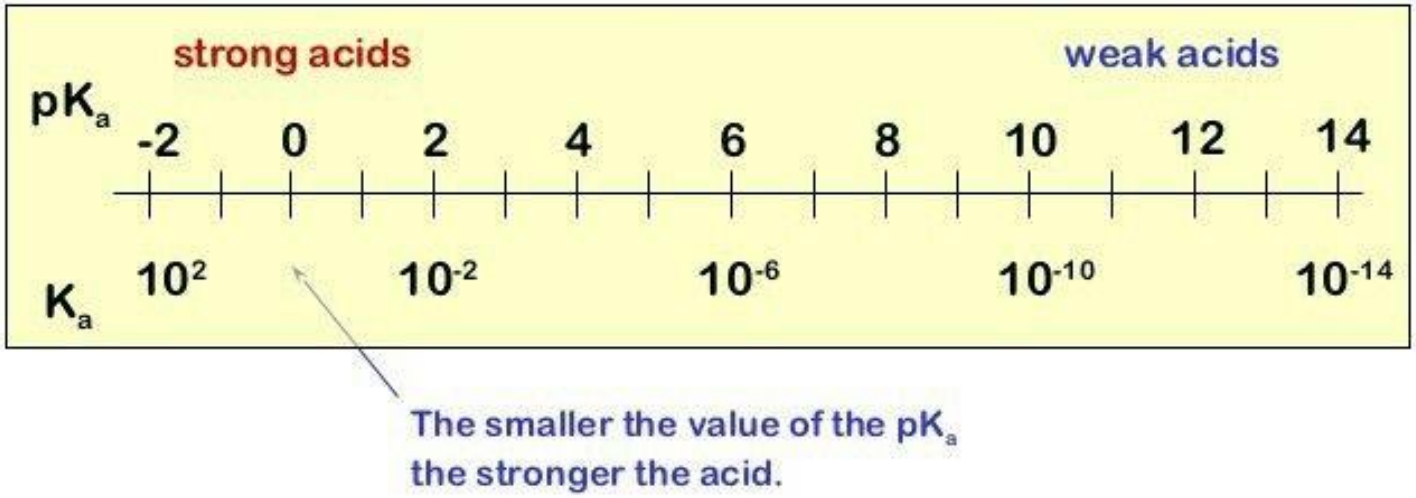
- Strength of an acid in water can be expressed by an acidity constant, K_a

$$K_a = \frac{[A^-][H_3O^+]}{[HA]} \qquad pK_a = -\log K_a$$

  - Smaller K_a and larger pK_a, less acidic
  - Larger K_a and smaller pK_a, more acidic
- Effect of alkyl substitution on alcohol acidity is primarily due to solvation of alkoxide ion formed on acid dissociation

COMPARISON OF pK_a and K_a VALUES

$$pK_a = -\log K_a$$



We will use pK_a to describe the strengths of acids.
It is a single number, without exponents.

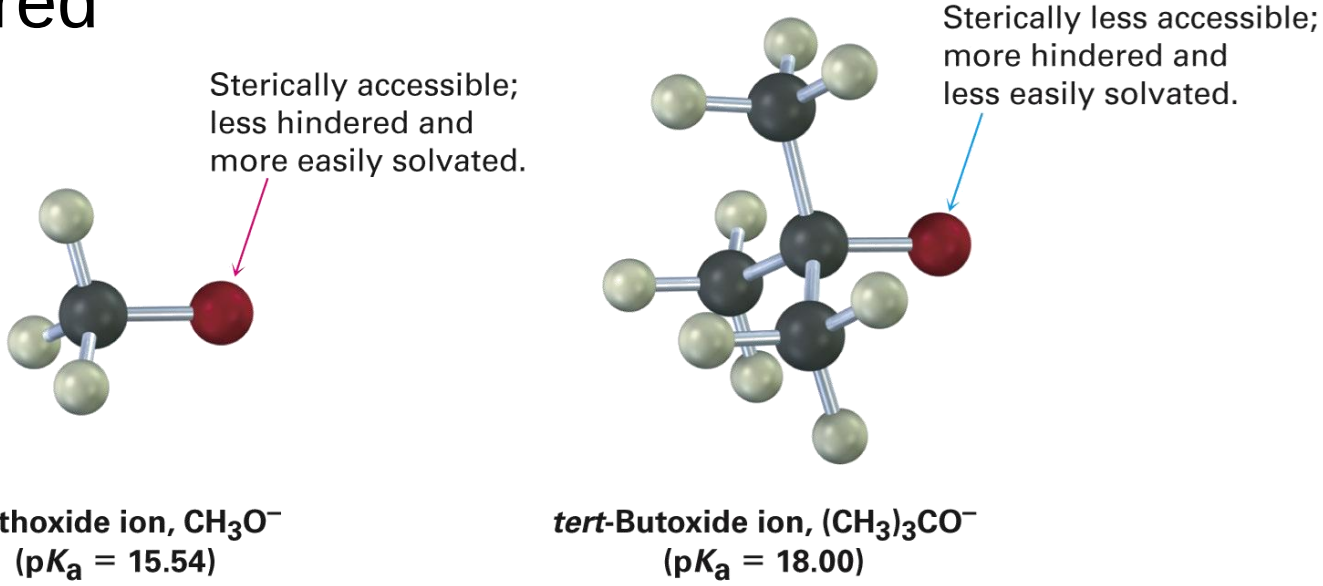
https://sk.pinterest.com/pin/209065607697034574/

Table 17.1 - Acidity Constants of Some Alcohols and Phenols

Compound	pK_a	
$(CH_3)_3COH$	18	Weaker acid
CH_3CH_2OH	16	
H_2O	15.74	
CH_3OH	15.54	
CF_3CH_2OH	12.43	
p-Aminophenol	10.46	
CH_3SH	10.3	
p-Methylphenol	10.17	
Phenol	9.89	
p-Chlorophenol	9.38	
p-Nitrophenol	7.15	Stronger acid

Properties of Alcohols and Phenols

- The more easily the alkoxide ion is solvated by water, the more its formation is energetically favored

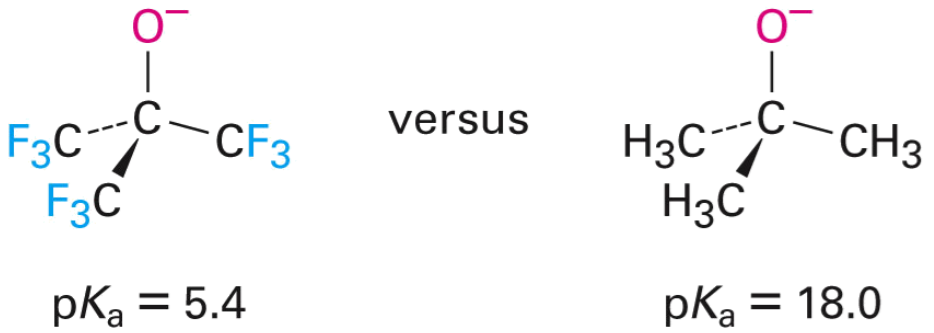


- Inductive effects are important in determining alcohol acidities

Properties of Alcohols and Phenols

- Electron-withdrawing groups make an alcohol a stronger acid by stabilizing the conjugate base

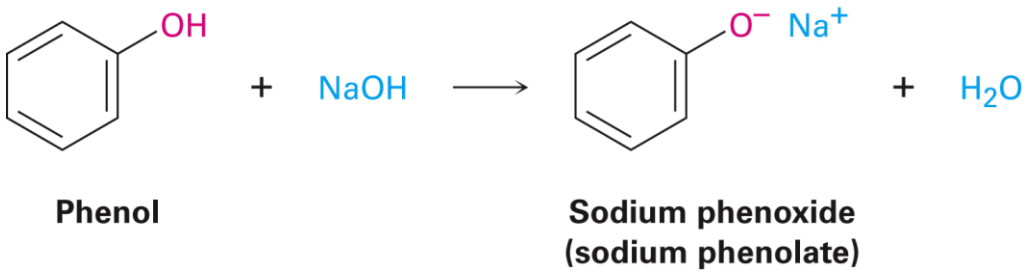
Electron-withdrawing groups stabilize the alkoxide ion and lower the pK_a.



pK_a = 5.4 pK_a = 18.0

- Alcohols react with alkali metals and with strong bases

Properties of Alcohols and Phenols



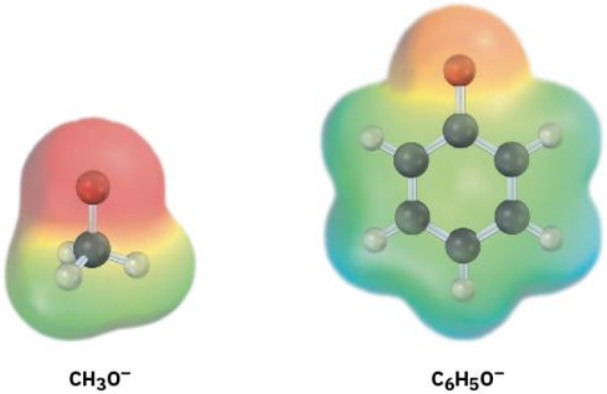
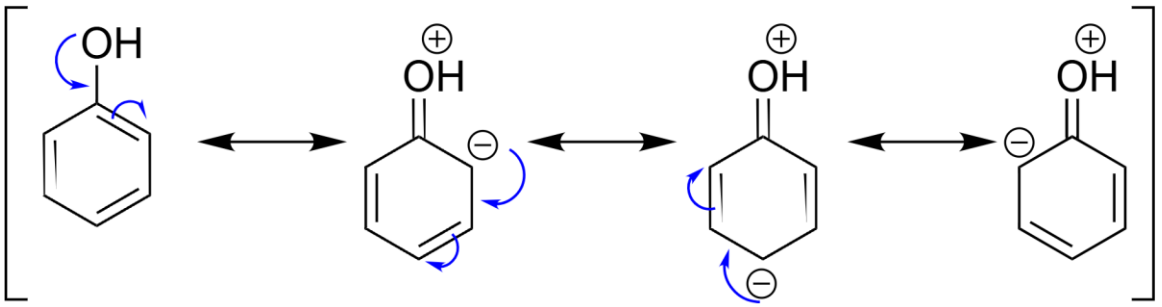
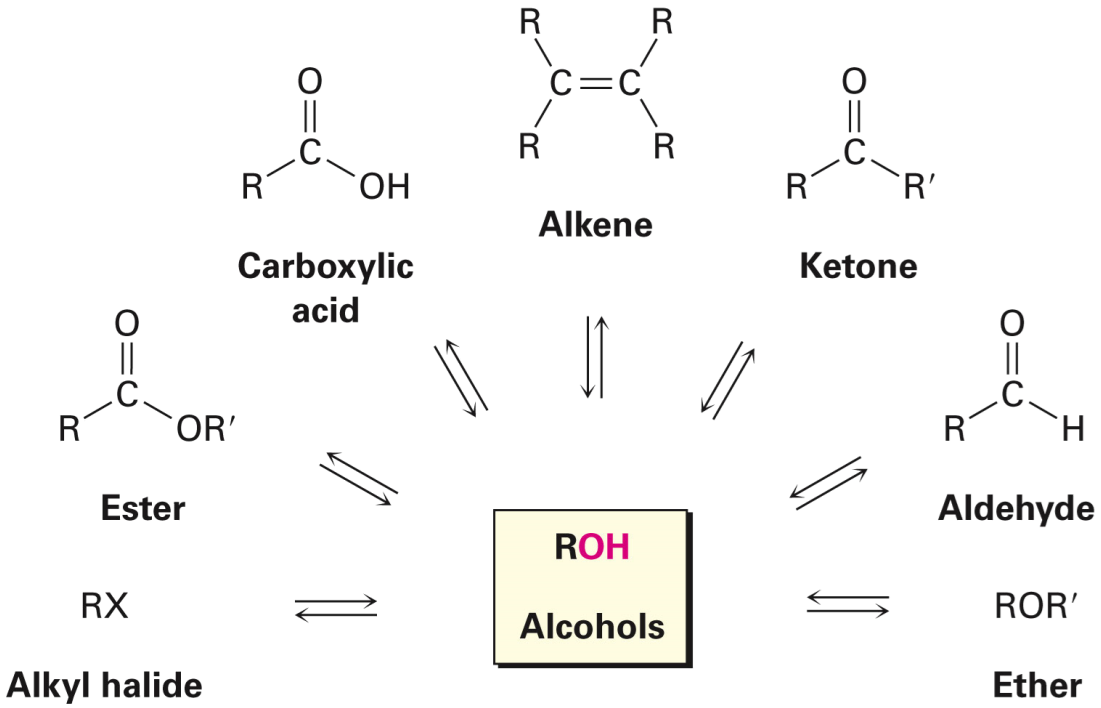


FIGURE 17.3 The resonance-stabilized phenoxide ion is more stable than an alkoxide ion



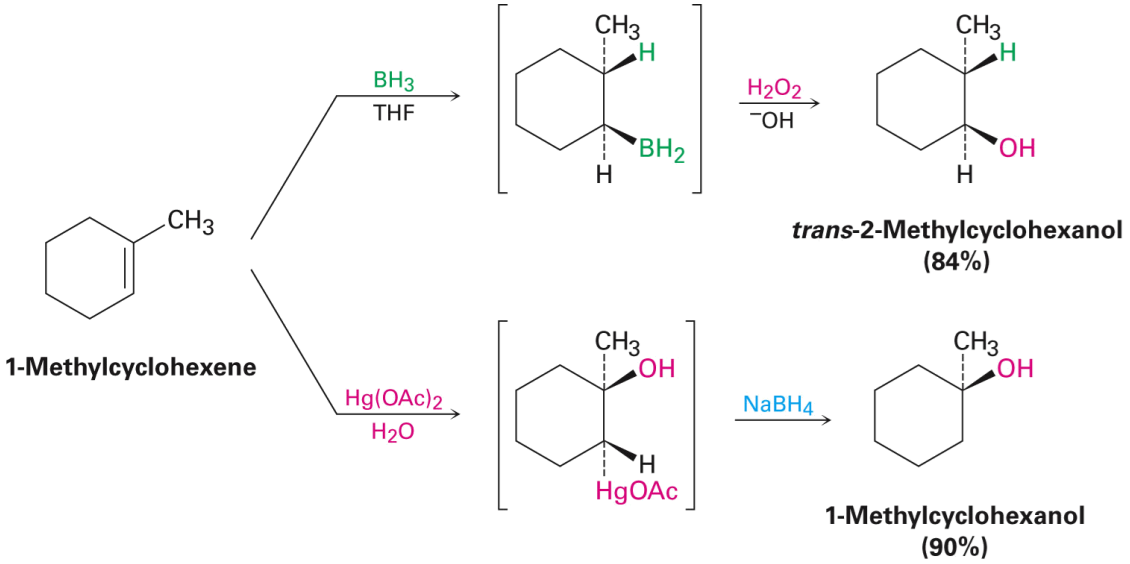
Preparation of Alcohols: A Review

- Alcohols are derived from many types of compounds
- The alcohol hydroxyl can be converted to many other functional groups



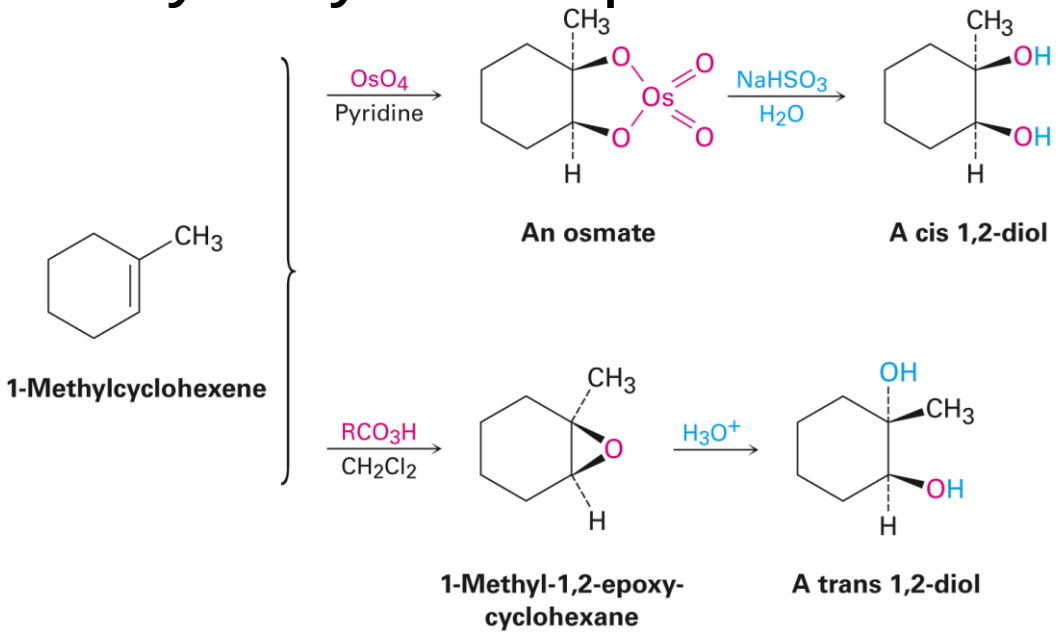
Preparation of Alcohols: A Review

- In the hydroboration of alkenes, indirect methods used are:
  - Hydroboration-oxidation, yielding the syn, non-Markovnikov hydration product
  - Oxymercuration-demercuration yielding Markovnikov hydration product



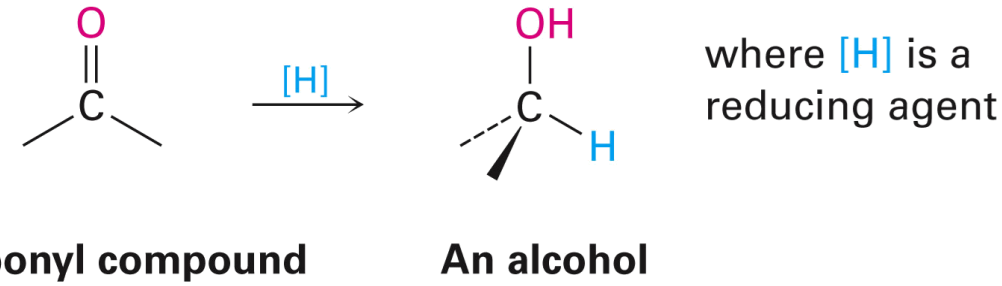
Preparation of Alcohols: A Review

- Cis-1,2-diols are prepared from direct hydroxylation of an alkene with OsO_4 followed by reduction with $NaHSO_3$
- Trans-1,2-diols can be prepared from acid-catalyzed hydrolysis of epoxides

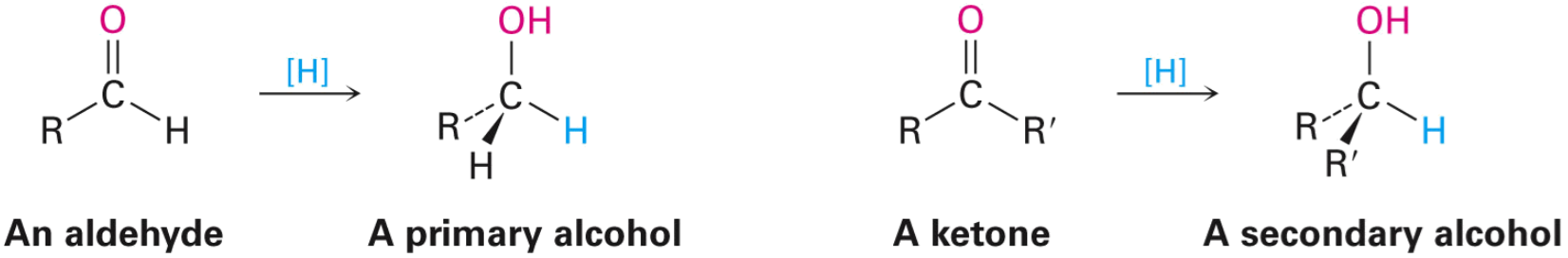


Alcohols from Carbonyl Compounds: Reduction

- Reduction of a carbonyl compound gives an alcohol
  - Addition of H to a C═O bond



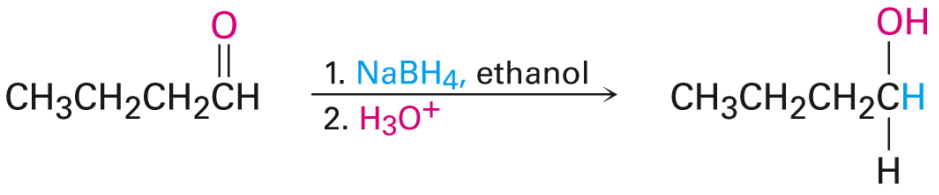
- Reduction of aldehydes gives primary alcohols
- Reduction of ketones gives secondary alcohols



Reduction Reagent: Sodium Borohydride ($NaBH_4$)

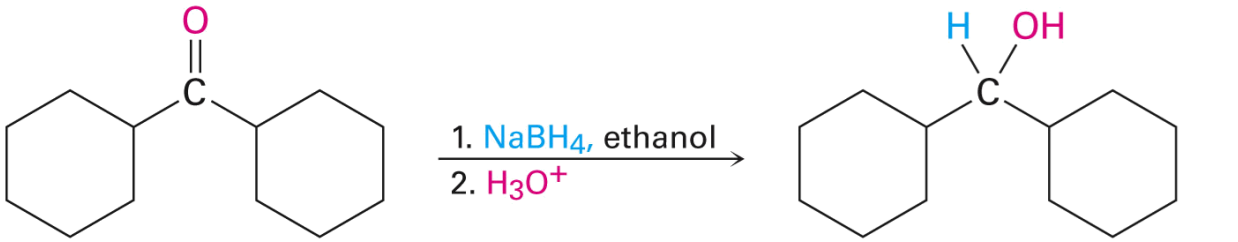
- $NaBH_4$ is not sensitive to moisture and can be used with either water or alcohol

Aldehyde reduction



Butanal — **1-Butanol (85%) (a 1° alcohol)**

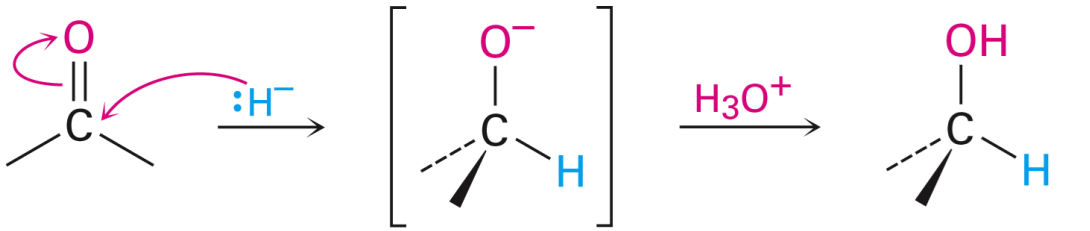
Ketone reduction



Dicyclohexyl ketone — **Dicyclohexylmethanol (88%) (a 2° alcohol)**

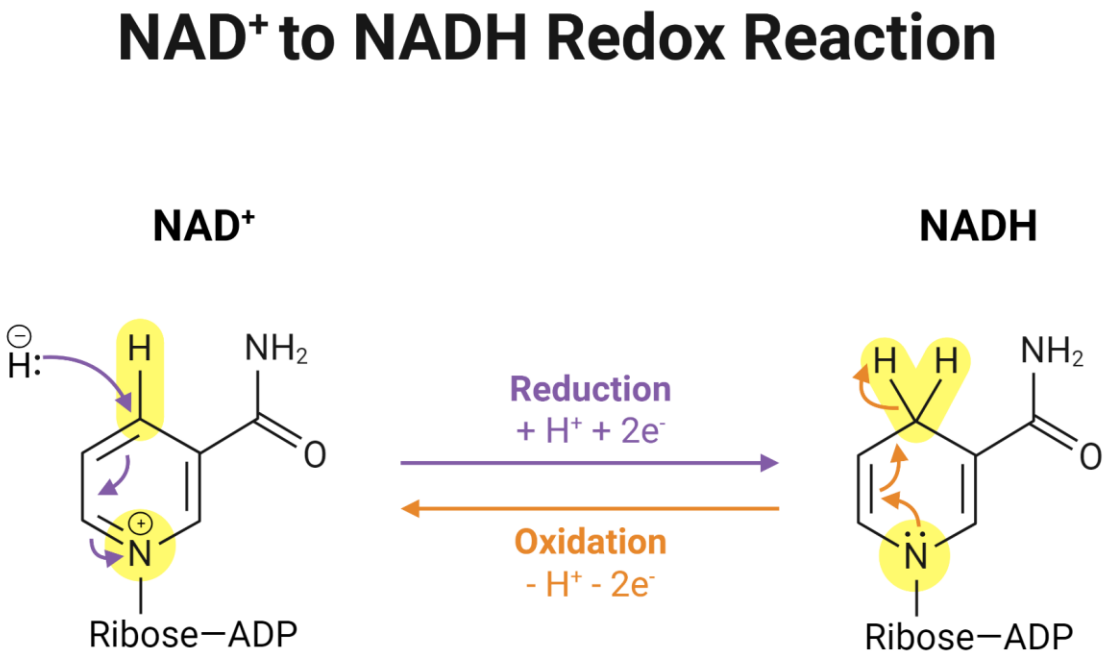
Mechanism of Reduction

- Addition of a nucleophilic hydride ion to the positively polarized, electrophilic carbon atom of the carbonyl group
- Alkoxide ion is protonated to yield the alcohol product



A carbonyl compound **An alkoxide ion intermediate** **An alcohol**

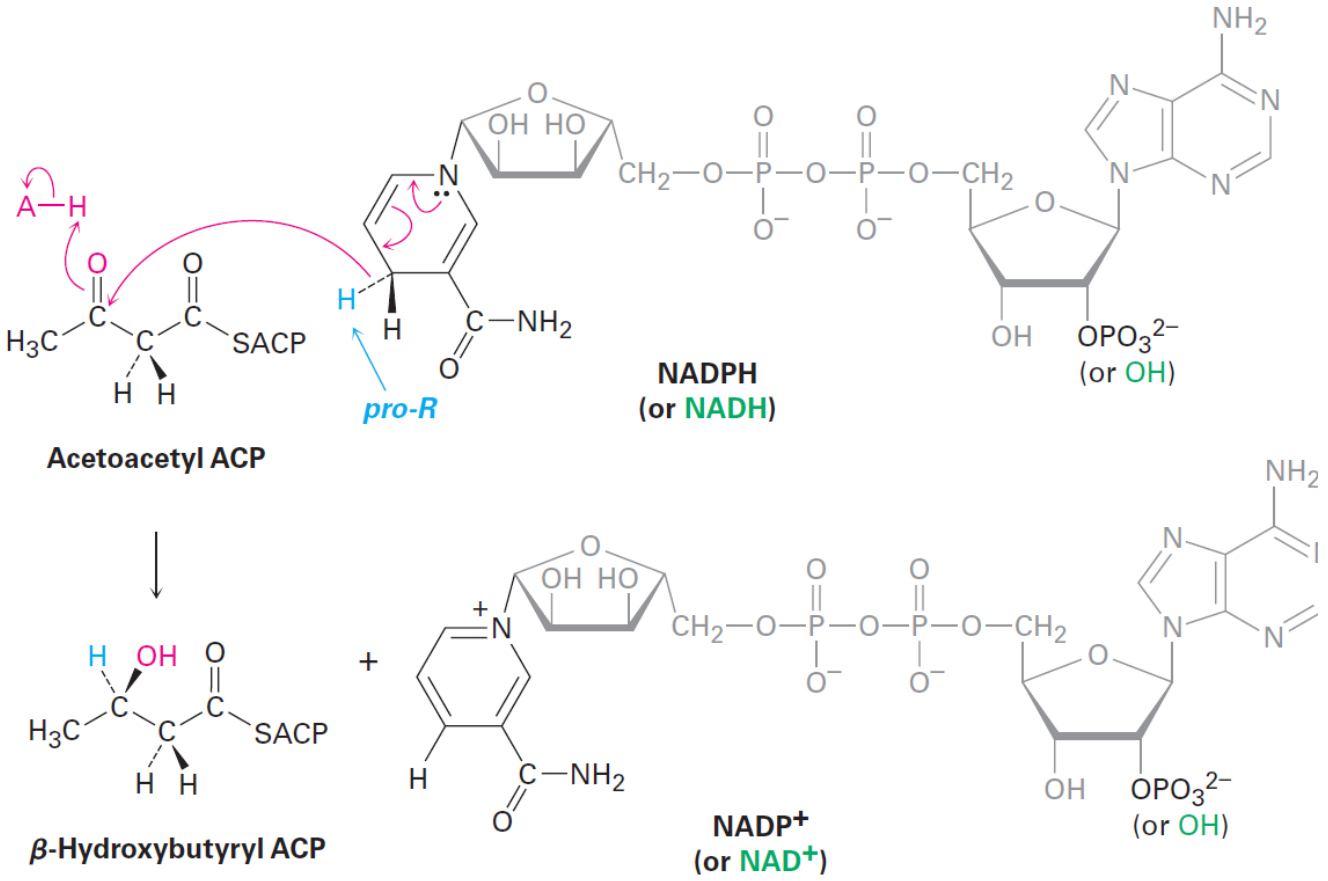
Example



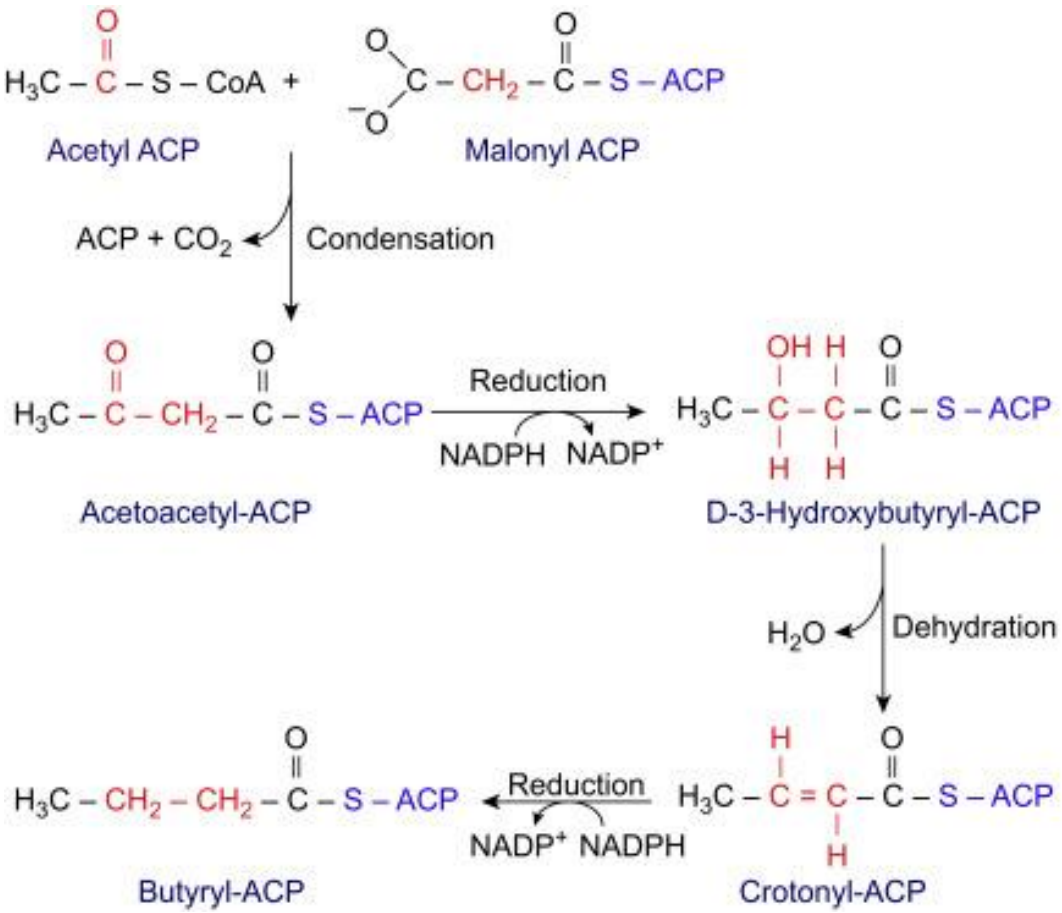
https://www.biorender.com/template/nad-to-nadh-redox-reaction

Reductions in Living Organisms

- Carried out by either of the coenzymes, NADH or NADPH



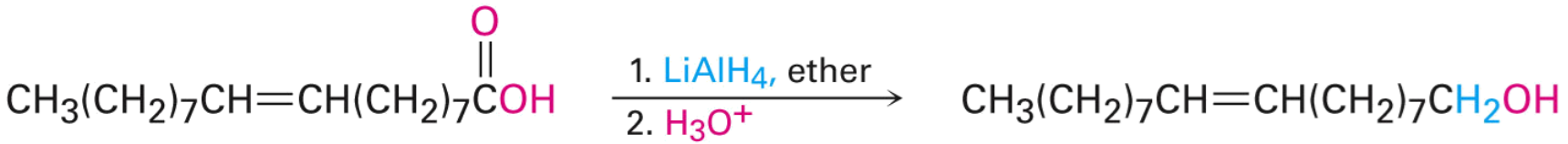
Reductions in Living Organisms



Reduction of Carboxylic Acids and Esters

- Carboxylic acids and esters are reduced to give primary alcohols
- $LiAlH_4$ is used because $NaBH_4$ is not effective

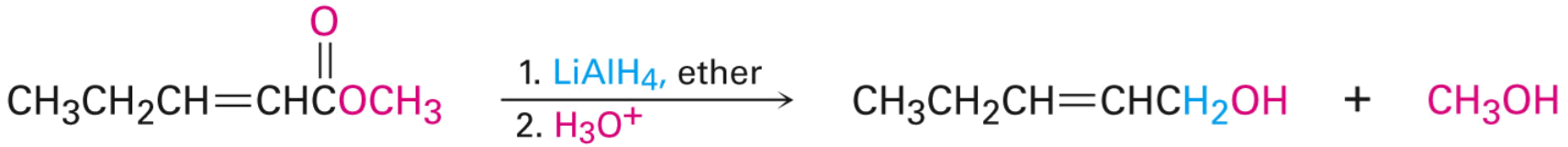
Carboxylic acid reduction



9-Octadecenoic acid (oleic acid)

9-Octadecen-1-ol (87%)

Ester reduction

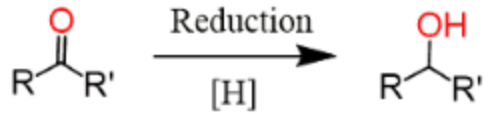


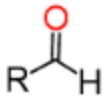
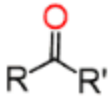
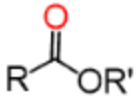
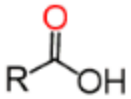
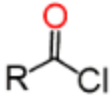
Methyl 2-pentenoate

2-Penten-1-ol (91%)

Example

Preparation of Alcohols by Reduction of Carbonyl Compounds



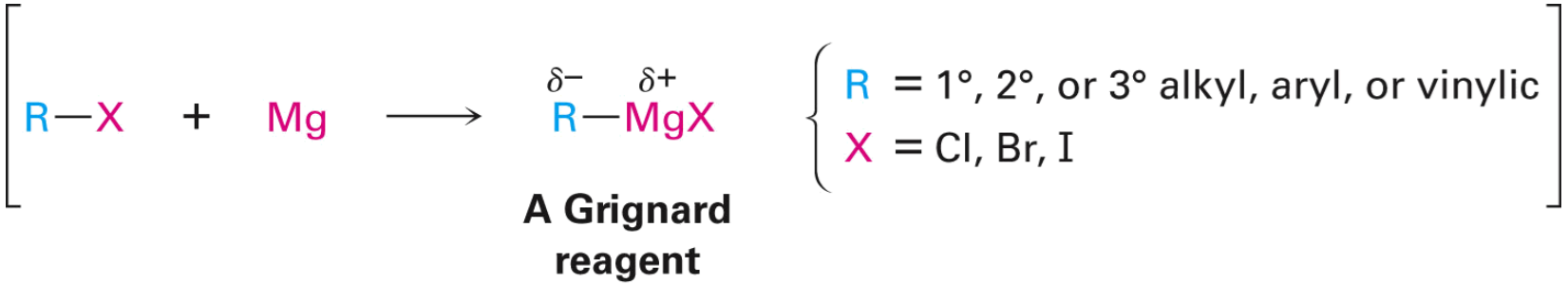
	[HH]	$LiAlH_4$	$NaBH_4$ (1 equiv)	Raney Ni	Pd/C	DIBAL-H
Aldehyde		✓	✓	✓	✓	✓
Ketone		✓	✓	✓	✓	✓
Ester		✓	✗	✗	Not effective	✓
Acid		✓	✗	✗		✓
Acid Chloride		✓	✓	✗		✓

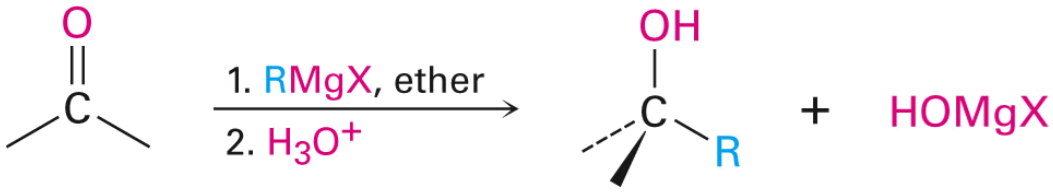
* DIBAL-H can reduce esters and acid chlorides to an aldehyde at -78 °C.

https://www.chemistrysteps.com/lialh4-and-nabh4-carbonyl-reduction-mechanism/

Alcohols from Carbonyl Compounds: Grignard Reagents

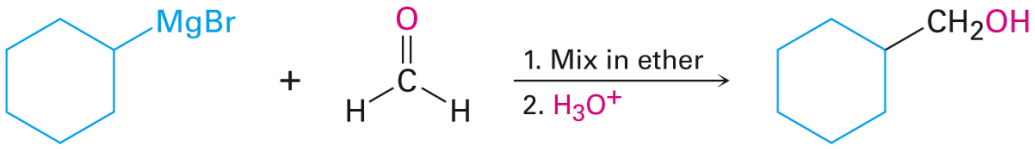
- Organohalides react with magnesium to produce Grignard reagents, RMgX
- Grignard reagents react with carbonyl compounds to yield alcohols





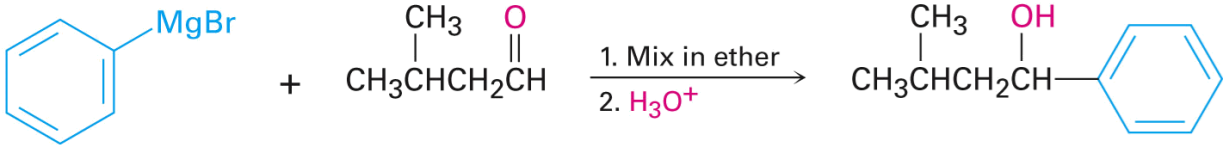
Alcohols from Carbonyl Compounds: Grignard Reagents

Formaldehyde reaction



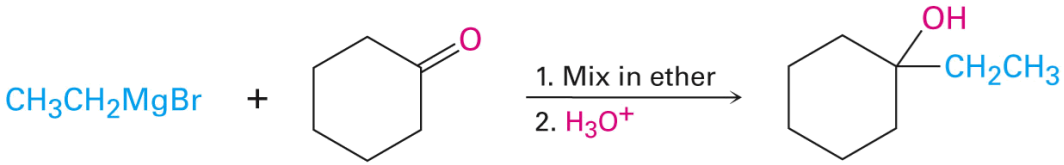
Cyclohexyl-magnesium bromide + **Formaldehyde** → **Cyclohexylmethanol (65%) (a 1° alcohol)**

Aldehyde reaction



Phenylmagnesium bromide + **3-Methylbutanal** → **3-Methyl-1-phenyl-1-butanol (73%) (a 2° alcohol)**

Ketone reaction



Ethylmagnesium bromide + **Cyclohexanone** → **1-Ethylcyclohexanol (89%) (a 3° alcohol)**

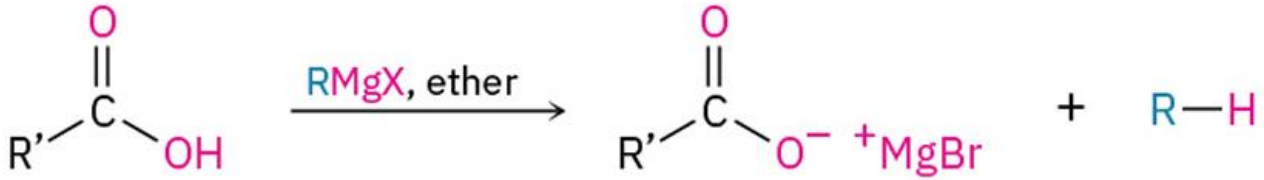
Alcohols from Carbonyl Compounds: Grignard Reagents



Ethyl pentanoate

2-Methyl-2-hexanol (85%)
(a 3° alcohol)



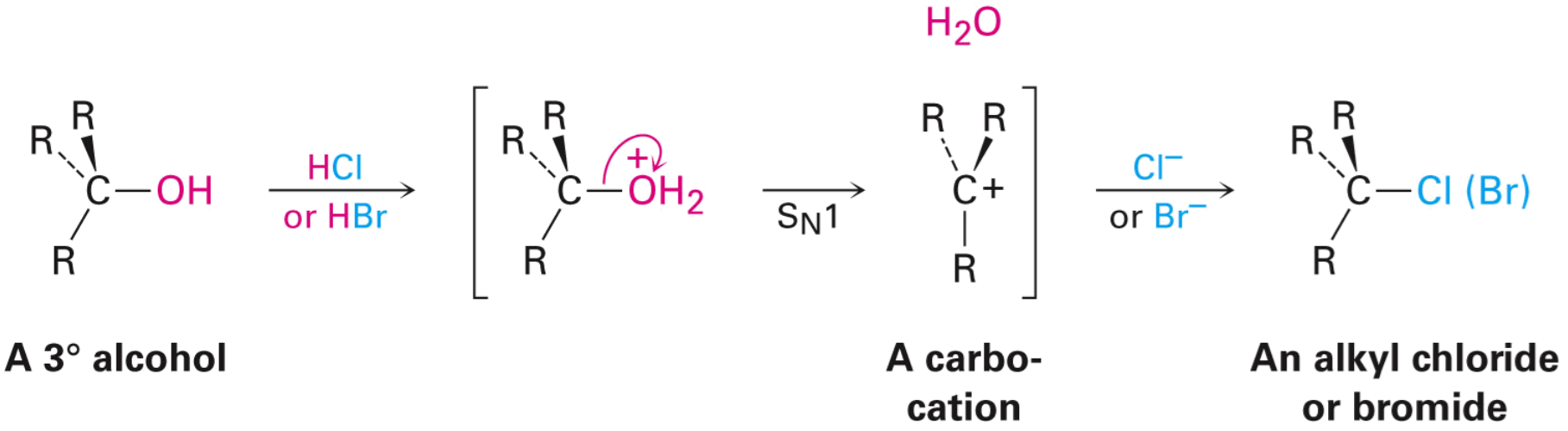


A carboxylic acid

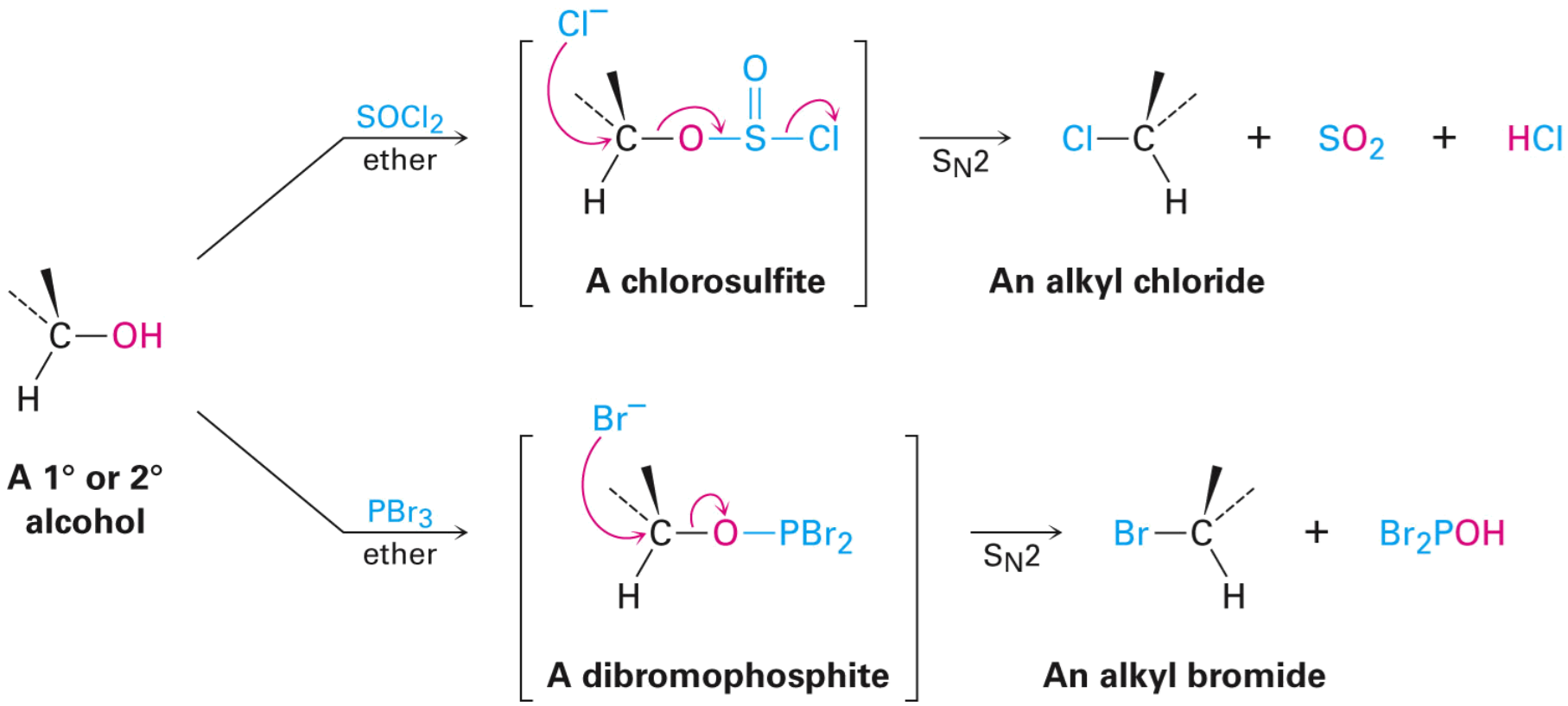
A carboxylic acid salt

Reactions of Alcohols

- Conversion of alcohols into alkyl halides
  - 3˚ alcohols react with HCl or HBr by S_N1 through carbocation intermediate
  - 1˚ and 2˚ alcohols are converted into halides by treatment with $SOCl_2$ or PBr_3 via S_N2 mechanism

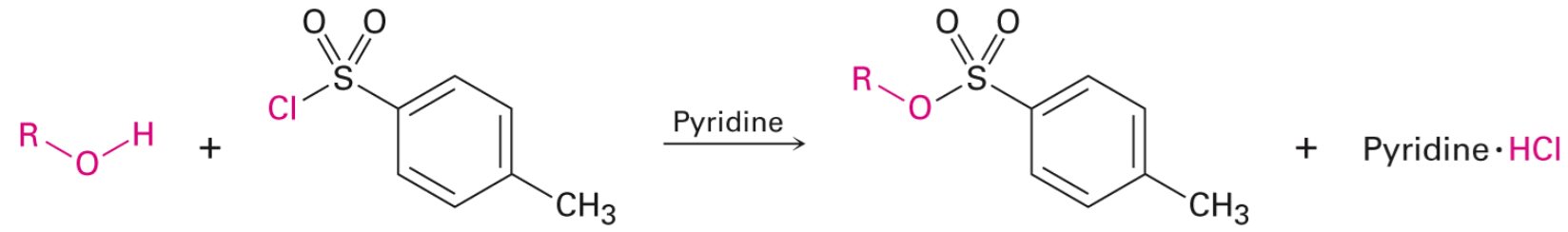


Reactions of 1˚ and 2˚ Alcohols



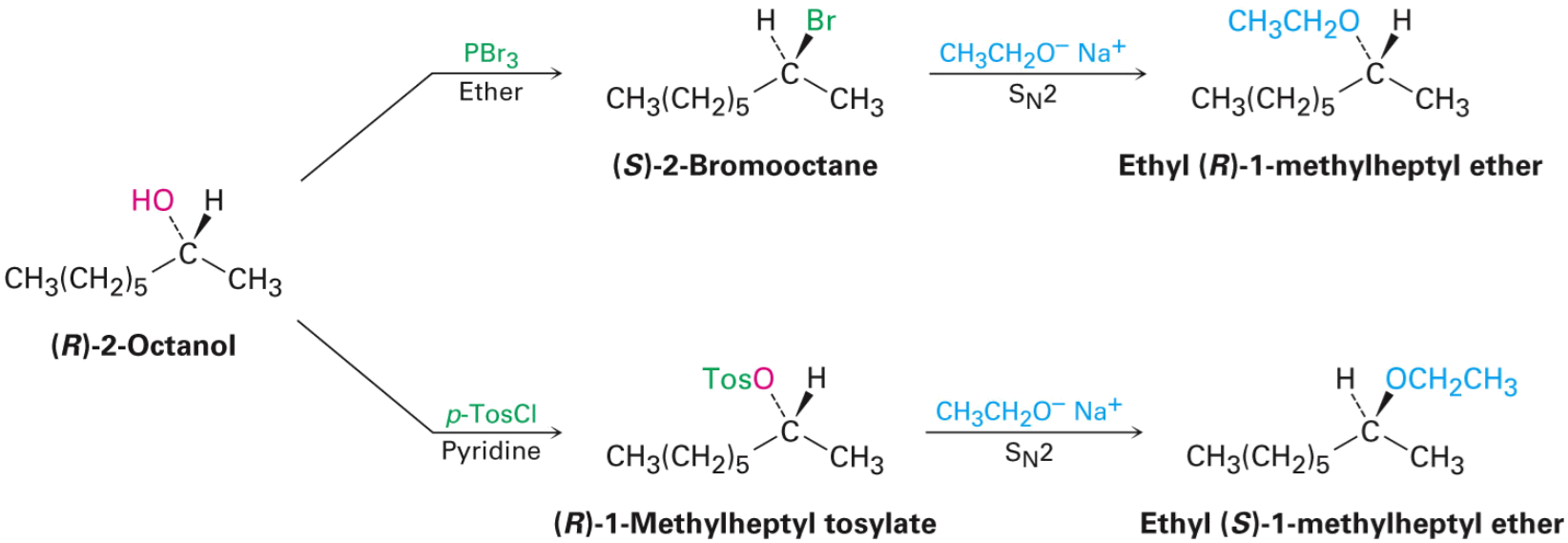
Conversion of Alcohols into Tosylates

- Reaction of alcohols with *p*-toluenesulfonyl chloride in pyridine yields alkyl tosylates, ROTos
- C–O bond remains intact and configuration at a chirality center is maintained
- Resulting alkyl tosylates react like alkyl halides



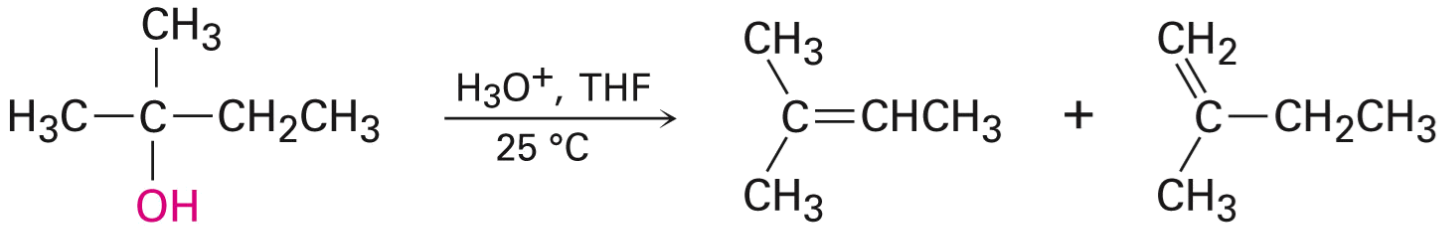
Stereochemical Uses of Tosylates

- The S_N2 reaction of an alcohol via an alkyl halide proceeds with two inversions, giving a product with same arrangement as the starting alcohol
- The S_N2 reaction of an alcohol via a tosylate, produces inversion at the chirality center



Dehydration of Alcohols to Yield Alkenes

- Tertiary alcohols are dehydrated using acid-catalyzed reactions
  - Follows Zaitsev's rule and yields a more stable alkene as the major product
- Reactivity is the result of the nature of the carbocation intermediate



2-Methyl-2-butanol

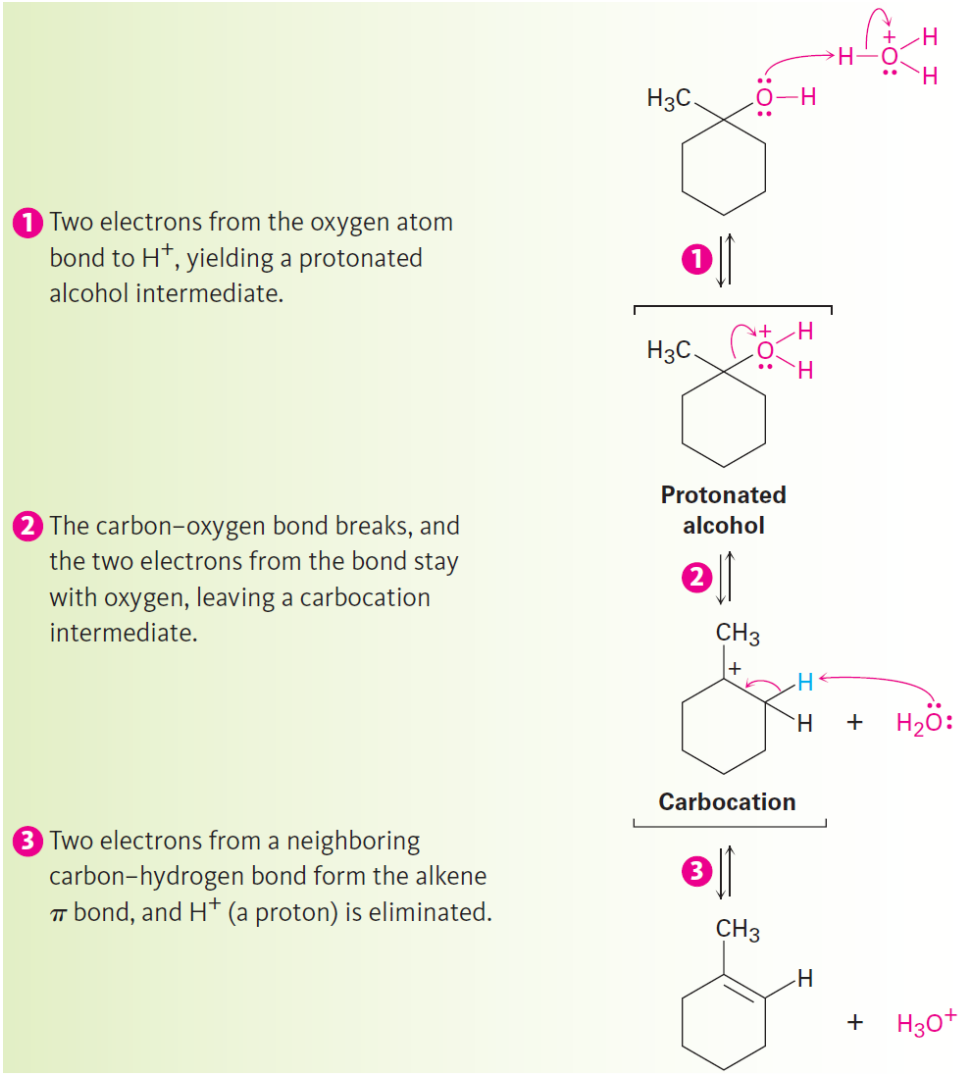
2-Methyl-2-butene (trisubstituted)

Major product

2-Methyl-1-butene (disubstituted)

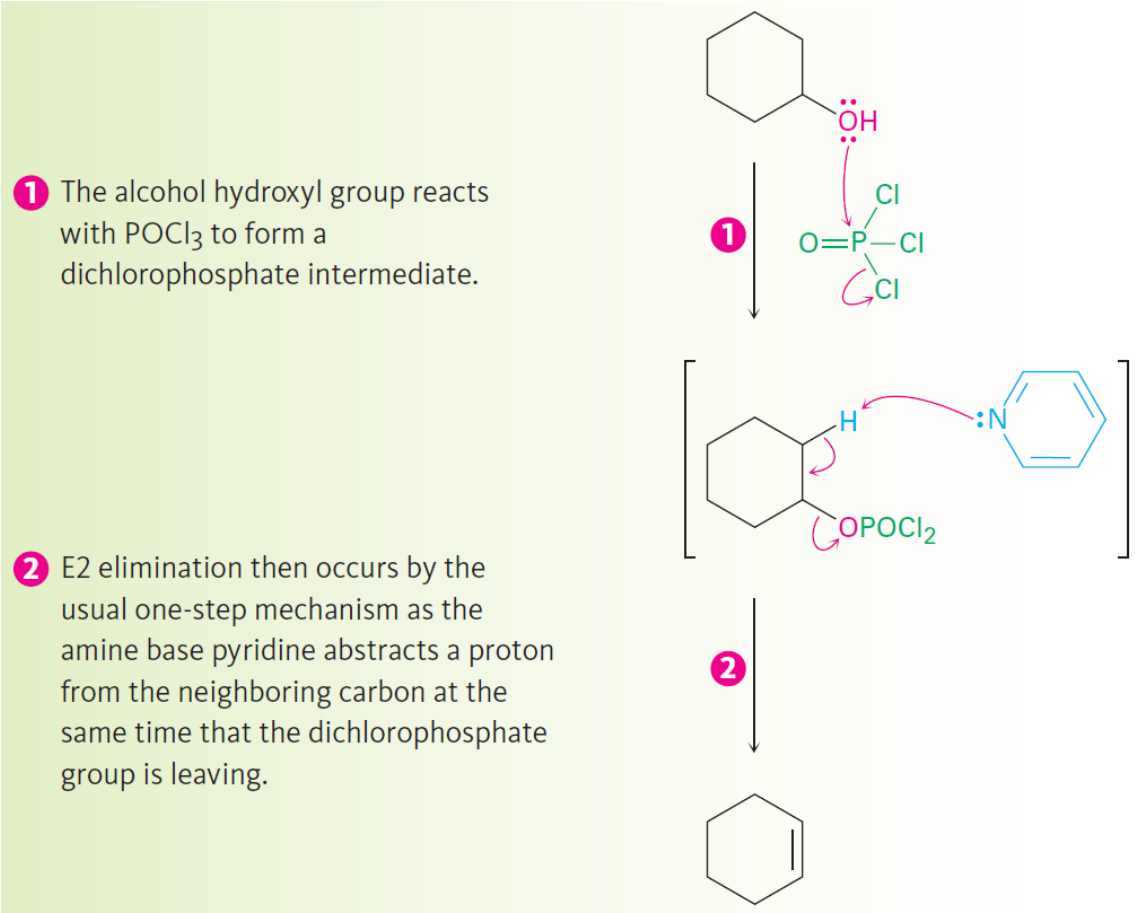
Minor product

Figure 17.6 - Mechanism



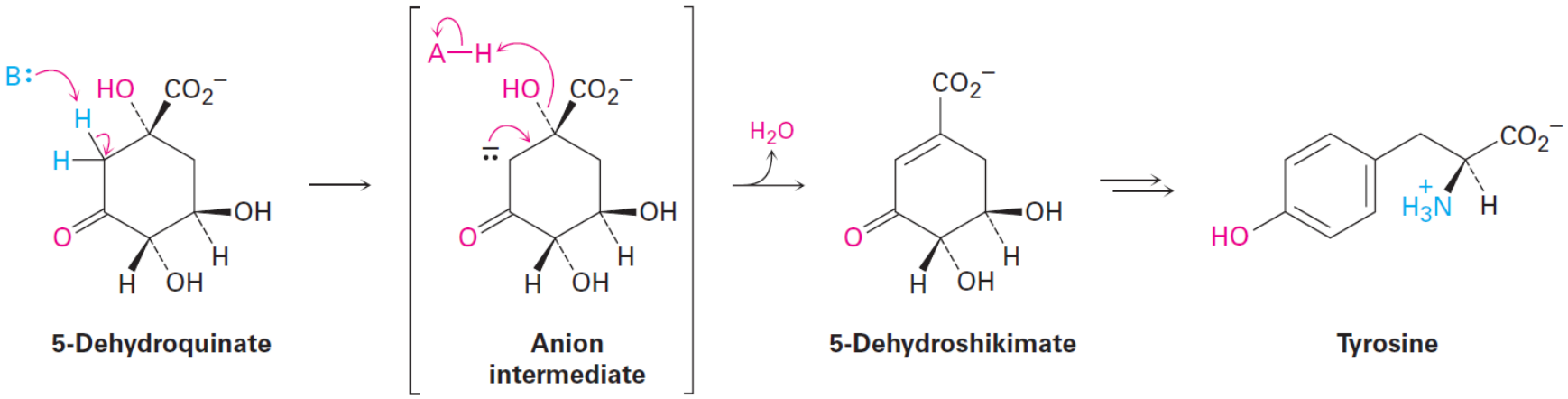
Dehydration with $POCl_3$

- Direct E2 elimination of water does not occur because hydroxide ion is a poor leaving group



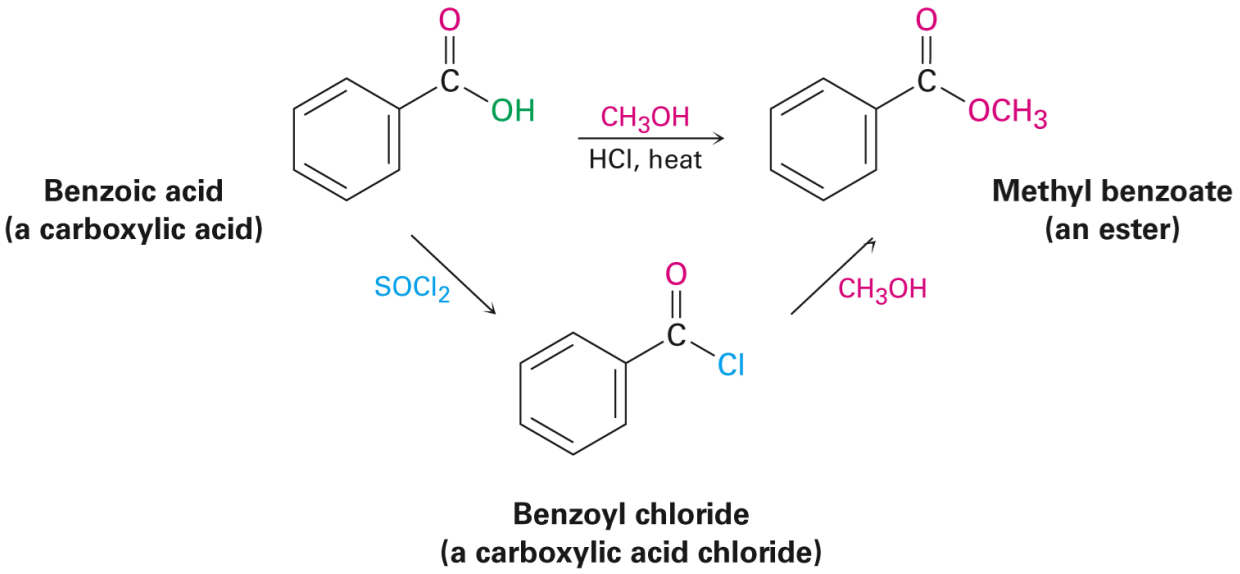
Biological Dehydrations

- Common and occur by an E1cB mechanism
- Example
  - Biosynthesis of the aromatic amino acid tyrosine



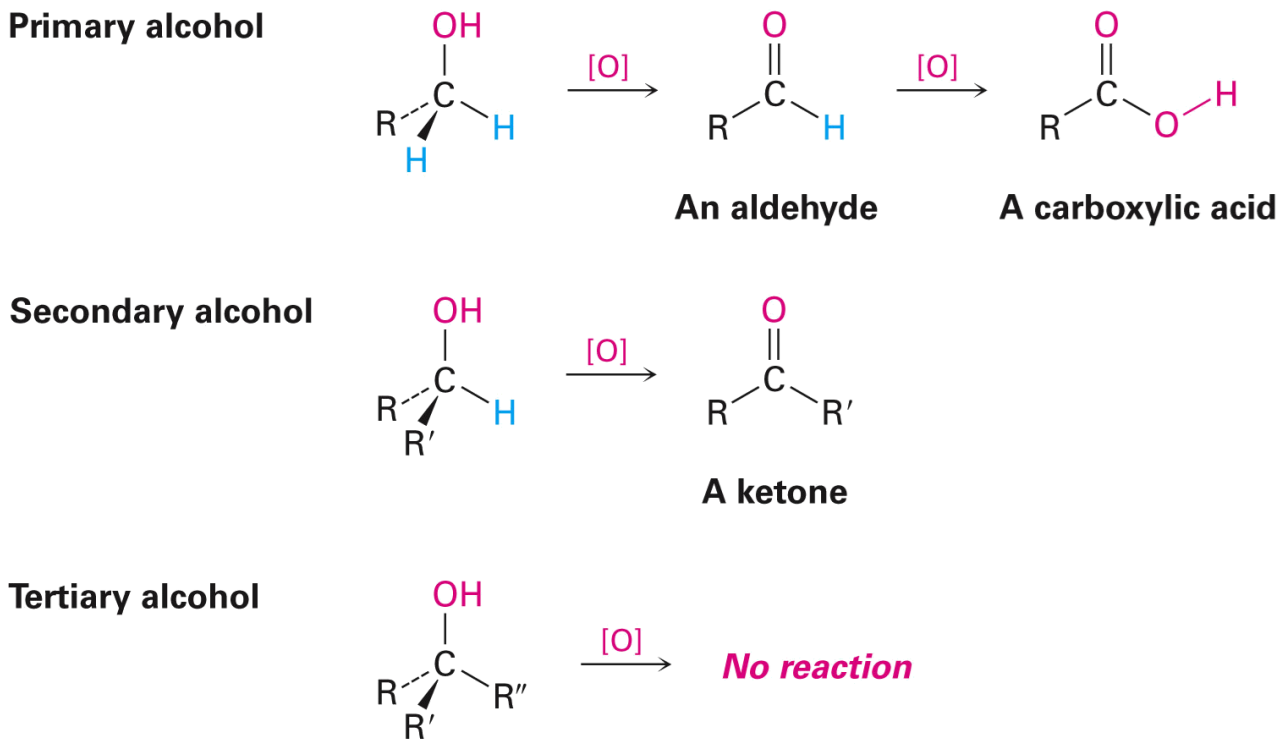
Conversion of Alcohols into Esters

- Reaction can be carried out in a single step with the use of a strong acid as catalyst
- Reactivity of carboxylic acid is increased by converting it into a carboxylic acid chloride, which then reacts with the alcohol



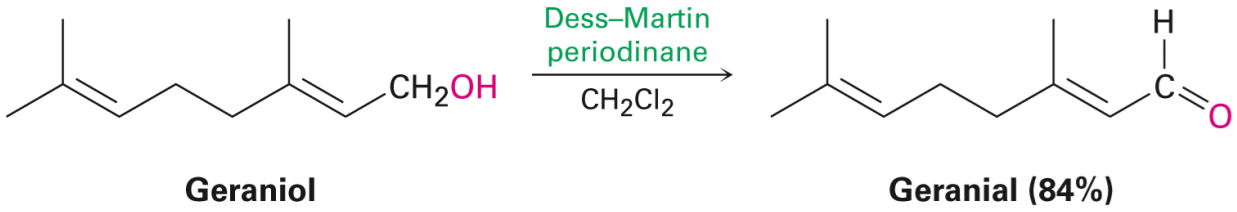
Oxidation of Alcohols

- Accomplished by reagents, such as $KMnO_4$, CrO_3, and $Na_2Cr_2O_7$

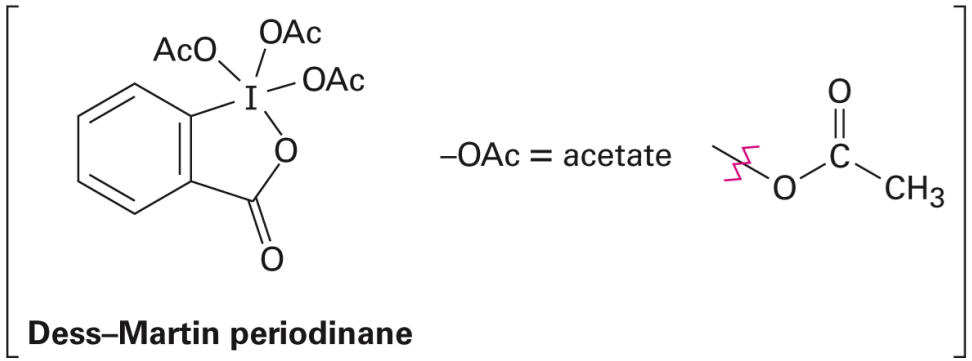


Oxidation of Alcohols

- Primary alcohols are oxidized to either aldehydes or carboxylic acids
- To prepare aldehyde from a primary alcohol, I(V)-containing Dess–Martin periodinane in dichloromethane is used

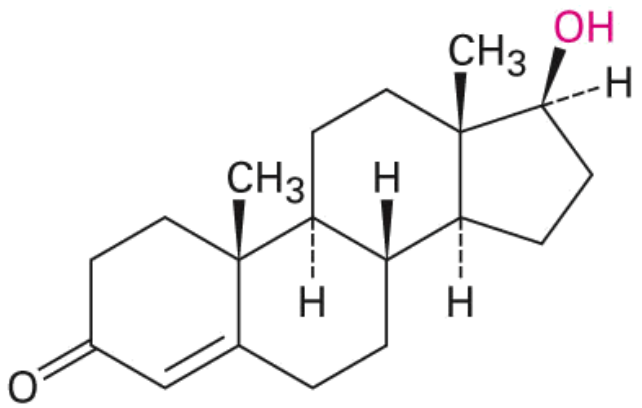


Geraniol **Geranial (84%)**



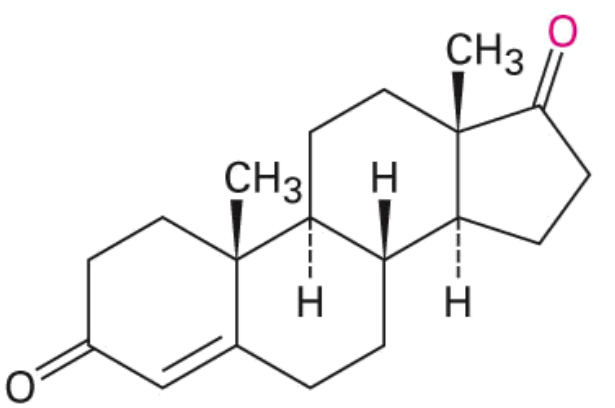
Dess–Martin periodinane

Oxidation of Alcohols



Testosterone
(male sex hormone)

PCC
CH_2Cl_2, 25 °C



4-Androstene-3,17-dione (82%)

$CH_3(CH_2)_8CH_2OH$

1-Decanol

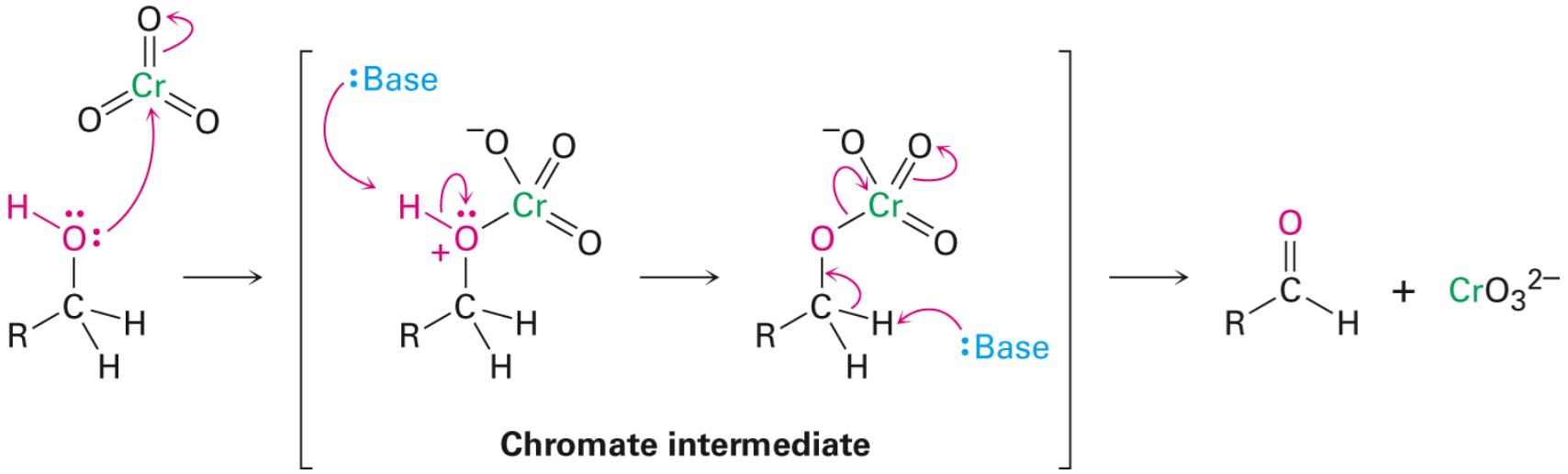
$KMnO_4/H_2O$
NaOH



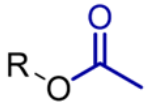
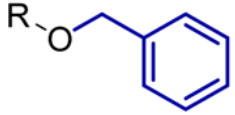
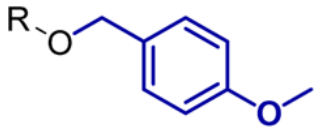
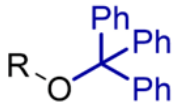
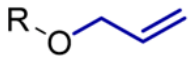
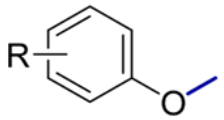
Decanoic acid (93%)

Oxidation of Alcohols

- Cr(VI) reagent reacts with the alcohol to give a chromate intermediate followed by expulsion of a reduced Cr(IV) species



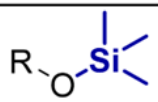
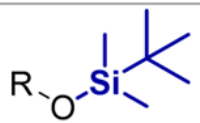
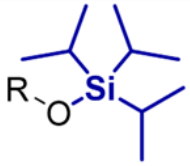
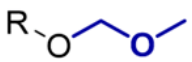
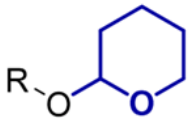
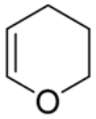
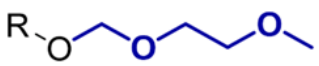
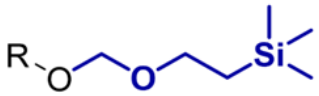
Alcohol Protecting Groups in Organic Chemistry

Class	Protecting group	Structure	Introduction	Deprotection
Ether	**Ac** Acetate		Ac_2O OR Ac-Cl + base (Et_3N, NaH, ...)	OH^- (K_2CO_3, KOH, ...)
	Bn Benzyl		BnBr/Cl + base (NaH, ...)	cat. Pd/C, H_2 (catalytic hydrogenolysis)
	PMB Para-methoxybenzyl		PMB-Br/Cl + base (NaH, ...)	a) DDQ, CAN (oxidants) b) cat. Pd/C, H_2
	Tr or Trt Trityl / triphenylmethyl		Ph_3CCl + cat. DMAP (in pyridine)	a) H^+, H_2O (AcOH, HCO_2H or stronger acids) b) $ZnBr_2$ (Lewis acids)
	All Allyl		Allyl bromide + base (NaH, ...)	cat. Pd(0), nucleophile (e.g., amines like morpholine)
	Me Methyl		MeI + base (NaH, ...)	BBr_3 or TMSI (Lewis acids)

Color: Base | acid (not all strictly need H_2O, can be anhydrous) | fluoride | Lewis acid | transition metal catalysis | oxidant

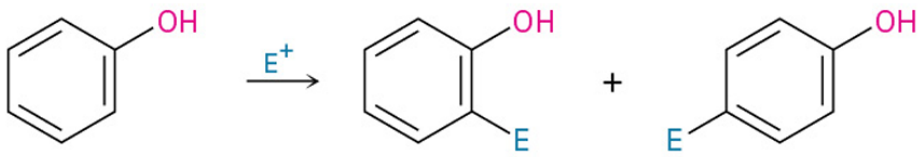
https://total-synthesis.com/alcohol-protecting-groups-in-organic-chemistry/

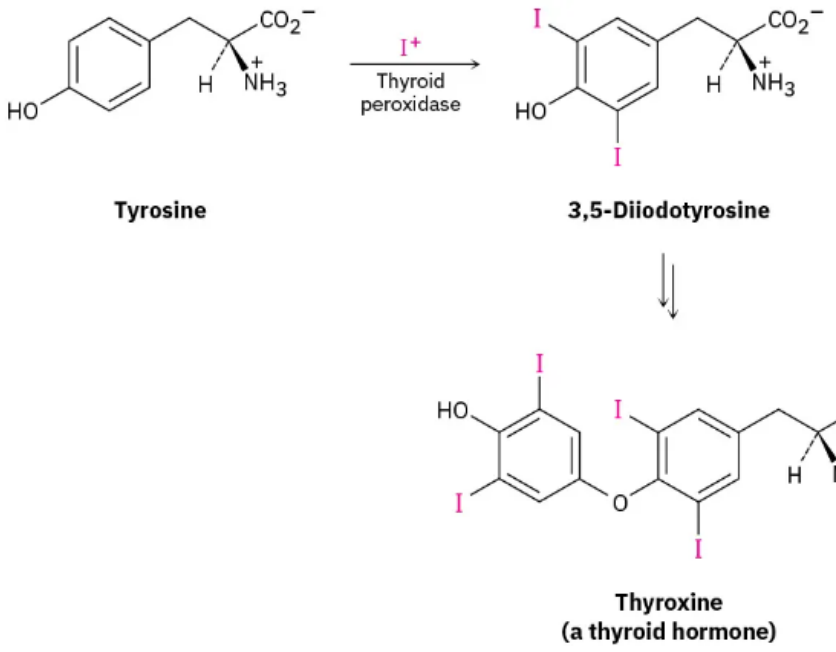
Alcohol Protecting Groups in Organic Chemistry

Class	Protecting group	Structure	Introduction	Deprotection
Silyl ether	**TMS** **Tri**methyl**s**ilyl		Silyl chloride + base (imidazole, pyridine, iPr$_2$NEt, NEt$_3$...) Alternatively, silyl triflate (more reactive)	a) F$^-$ (TBAF, HF...) b) H$^+$, H$_2$O (PPTS, ...) c) OH$^-$ (KOH, ...) TBS: As TMS, but much more stable to base TIPS: As TBS, but more stable to acid and base; needs harsher conditions
	TBS/TBDMS *t*-**B**utyl**dim**ethyl**s**ilyl			
	TIPS **Tri**i**so**propyl**s**ilyl			
Acetal	**MOM** **M**eth**o**xy**m**ethyl		MOM-Cl + base (iPr$_2$NEt, NaH, ...)	H$^+$, H$_2$O (HCl, PPTS, ...)
	THP **T**etra**h**ydro**p**yranyl		 (DHP) + H$^+$	H$^+$, H$_2$O (HCl, PPTS, ...)
	MEM 2-**M**ethoxy**e**thoxy**m**ethyl		MEM-Cl + base (iPr$_2$NEt, NaH, ...)	a) ZnBr$_2$ (Lewis acids) Chelation by 2nd O increases Lewis acid instability b) H$^+$, H$_2$O (TFA, PPTS)
	SEM 2-(Trimethyl**s**ilyl)**e**thoxy**m**ethyl		SEM-Cl + base (iPr$_2$NEt, NaH, ...)	a) F$^-$ (TBAF, HF, ...) b) H$^+$, H$_2$O (TFA, PPTS, ...)

Color: Base | acid (not all strictly need H$_2$O, can be anhydrous) | fluoride | Lewis acid | transition metal catalysis | oxidant

Reactions of Phenols



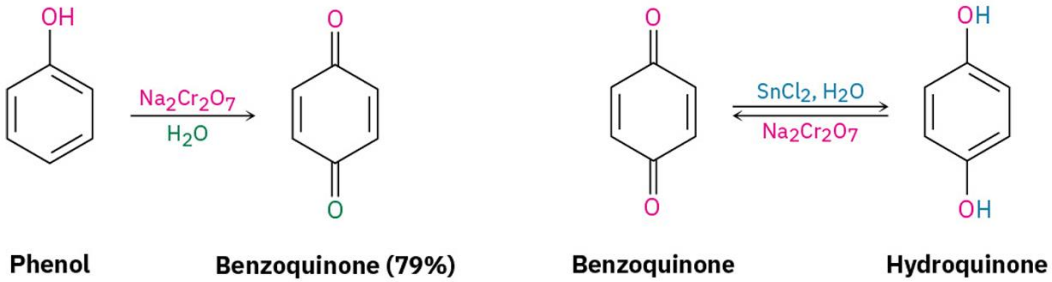


Thyroid hormones (THs) regulate metabolic processes important for normal growth and development and play essential regulatory roles in metabolism

The LibreTexts libraries
(16.2: Other Aromatic Substitutions)

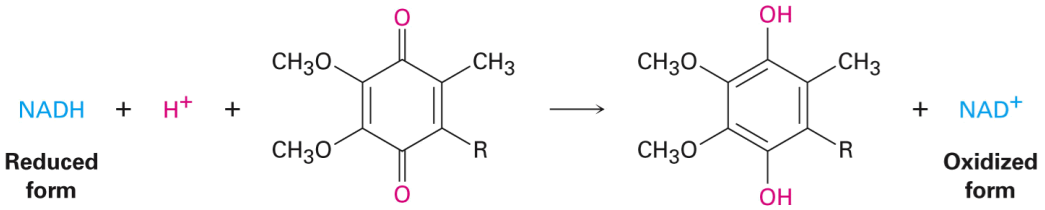
Brent, G.A. Mechanisms of thyroid hormone action. *J. Clin. Investig.* **2012**, *122*, 3035–3043.

Reactions of Phenols

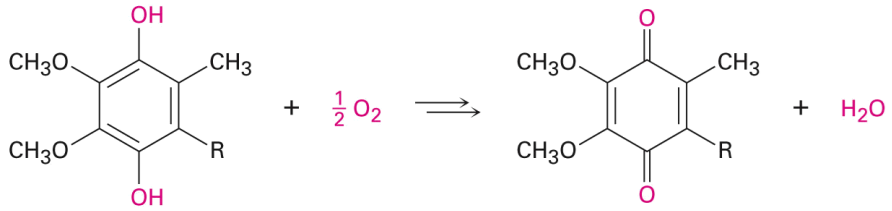


- Ubiquinones mediate electron-transfer processes involved in energy production

Step 1



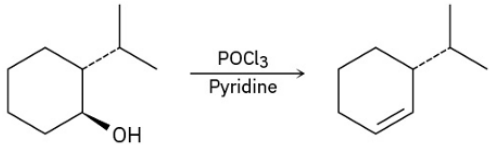
Step 2



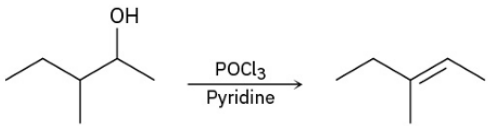
Net change: NADH + $\frac{1}{2}O_2$ + H^+ ⟶ NAD^+ + H_2O

PROBLEM 17-33 For each reaction, write the mechanism using curved arrows for the conversion of the alcohol into the corresponding alkene with $POCl_3$. In each case, explain the regiochemistry of the elimination.

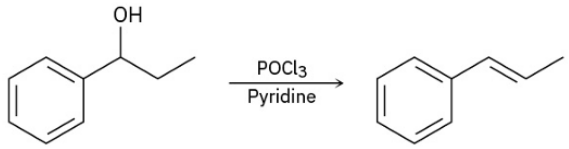
(a)



(b)

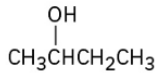


(c)

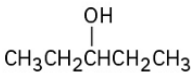


PROBLEM 17-42 What Grignard reagent and what carbonyl compound might you start with to prepare the following alcohols:

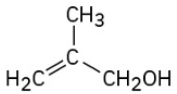
(a)



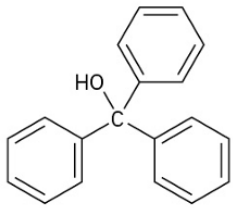
(b)



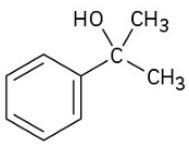
(c)



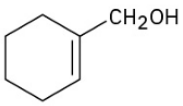
(d)



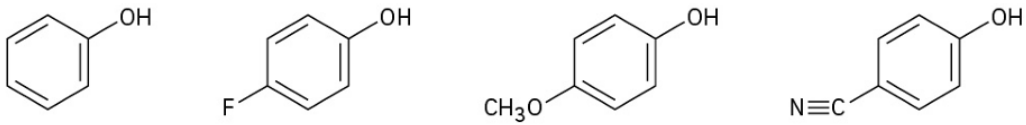
(e)



(f)



PROBLEM 17-59 Rank the following substituted phenols in order of increasing acidity, and explain your answer:



PROBLEM 17-72 Identify the reagents **a**–**f** in the following scheme:

